Power Series and Interval of Convergence

ANSWER KEY

Section 1: Finding the Radius of Convergence

1 Find the radius of convergence R for the power series $\sum_{n=0}^{\infty} \frac{(x-3)^n}{4^n}$.

a. $R = 4$ ✓

b. $R = 1$

c. $R = 3$

d. $R = \infty$

Ratio test: $\lim_{n\to\infty} \left|\frac{(x-3)^{n+1}}{4^{n+1}} \frac{4^n}{(x-3)^n}\right| = \frac{|x-3|}{4} < 1 \Rightarrow |x-3| < 4$. The radius is $R = 4$.

2 What is the radius of convergence of $\sum_{n=0}^{\infty} \frac{x^n}{n!}$?

a. $R = \infty$ ✓

b. $R = 1$

c. $R = 0$

d. $R = e$

Ratio test: $\lim_{n\to\infty} \left|\frac{x^{n+1}}{(n+1)!} \frac{n!}{x^n}\right| = \lim_{n\to\infty} \frac{|x|}{n+1} = 0 < 1$ for all real x. Radius $R = \infty$.

3 Find the radius of convergence of $\sum_{n=1}^{\infty} n!\, x^n$.

a. $R = 0$ **(converges only at $x = 0$)** ✓

b. $R = 1$

c. $R = \infty$

d. $R = \frac{1}{e}$

Ratio test: $\lim_{n\to\infty} (n+1)|x| = \infty$ for any $x \neq 0$. The series converges only at its center $x = 0$, so $R = 0$.

Section 2: Testing Interval Endpoints

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

4 Find the complete interval of convergence for $\sum_{n=1}^{\infty} \frac{(x-2)^n}{n}$.

a. $[1, 3)$ ✓

b. $(1, 3)$

c. $[1, 3]$

d. $(1, 3]$

Ratio test gives radius $R = 1$, open interval $(1, 3)$. At left endpoint $x = 1$: $\sum \frac{(-1)^n}{n}$ converges by AST. At right endpoint $x = 3$: $\sum \frac{1}{n}$ diverges (harmonic). Interval is $[1, 3)$.

5 What is the interval of convergence of $\sum_{n=1}^{\infty} \frac{x^n}{n^2}$?

a. $[-1, 1]$ ✓

b. $(-1, 1)$

c. $[-1, 1)$

d. $(-1, 1]$

Radius $R = 1$. At $x = 1$, $\sum \frac{1}{n^2}$ converges ($p = 2$). At $x = -1$, $\sum \frac{(-1)^n}{n^2}$ converges absolutely. Both endpoints are included: $[-1, 1]$.

6 Why must endpoints always be tested individually when finding the interval of convergence?

a. **The Ratio Test yields $L = 1$ at the boundary points $|x - c| = R$ and is inconclusive** ✓

b. Endpoints always diverge

c. Endpoints always converge

d. To find the center of the series

Because the Ratio Test limit equals 1 at $x = c \pm R$, it yields no information at endpoints, necessitating individual convergence tests.

Section 3: Term-by-Term Differentiation of Power Series

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

7 If $f(x) = \sum_{n=0}^{\infty} c_n x^n$ has radius of convergence R, what is the radius of convergence of its derivative $f'(x) = \sum_{n=1} n c_n x^{n-1}$?

a. R **(the radius is unchanged)** ✓

b. $R - 1$

c. $\frac{R}{2}$

d. ∞

A fundamental property of power series is that term-by-term differentiation and integration preserve the exact same radius of convergence R.

8 Starting from $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$, find a power series representation for $\frac{1}{(1-x)^2}$.

a. $\sum_{n=1}^{\infty} n x^{n-1}$ ✓

b. $\sum_{n=0}^{\infty} x^{2n}$

c. $\sum_{n=0}^{\infty} \frac{x^{n+1}}{n+1}$

d. $\sum_{n=1}^{\infty} n^2 x^n$

Differentiate both sides: $\frac{d}{dx}\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$. Differentiating the series term-by-term gives $\sum_{n=1}^{\infty} n x^{n-1}$.

9 Can endpoint convergence change when a power series is differentiated term-by-term?

a. **Yes, differentiation can lose convergence at an endpoint** ✓

b. No, endpoints never change

c. Differentiation always adds endpoints

d. Only if $R = \infty$

While the radius R remains identical, differentiation may cause a previously convergent endpoint to become divergent.

Section 4: Term-by-Term Integration of Power Series

© ClipSAT: By Mr. Mohamed Abdallah +966597688647

10 Integrating $\frac{1}{1+x} = \sum_{n=0}^{\infty} (-1)^n x^n$ term-by-term yields the Maclaurin series for which function?

a. $\ln(1+x)$ ✓

b. $\arctan x$

c. e^{-x}

d. $\frac{1}{(1+x)^2}$

$\int_0^x \frac{1}{1+t} dt = \ln(1+x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \ \ldots$

11 Integrating $\frac{1}{1+x^2} = \sum_{n=0}^{\infty} (-1)^n x^{2n}$ term-by-term yields the Maclaurin series for:

a. $\arctan x$ ✓

b. $\ln(1+x^2)$

c. $\arcsin x$

d. $\cos x$

$\int_0^x \frac{1}{1+t^2} dt = \arctan x = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1} = x - \frac{x^3}{3} + \frac{x^5}{5} - \ \ldots$

12 Evaluate $\int_0^{0.1} \frac{1}{1+x^4} dx$ using the first two non-zero terms of its power series.

a. $0.1 - \frac{0.1^5}{5}$ ✓

b. $0.1 + \frac{0.1^5}{5}$

c. $0.1 - 0.1^4$

d. 0.1

Series: $1 - x^4 + x^8 - \ \ldots$ Integrating gives $[x - \frac{x^5}{5}]_0^{0.1} = 0.1 - \frac{(0.1)^5}{5} = 0.1 - 0.000002.$

© ClipSAT: By Mr. Mohamed Abdallah +966597688647